It can also be prepared by heating a metal nitrides such as magnesium nitride with water.

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

It is industrially manufactured by passing nitrogen and hydrogen over iron catalyst (a small amount of K_2O and Al_2O_3 is also used to increase the rate of attainment of equilibrium) at 750 K at 200 atm pressure. In the actual process the hydrogen required is obtained from water gas and nitrogen from fractional distillation of liquid air.

Properties

Ammonia is a pungent smelling gas and is lighter than air. It can be readily liquefied by at about 9 atmospheric pressure. The liquid boils at -38.4°C and freezes at -77° C. Liquid ammonia resembles water in its physical properties. i.e. it is highly associated through strong hydrogen bonding. Ammonia is extremely soluble in water (702 Volume in 1 Volume of water) at 20°C and 760mm pressure.

At low temperatures two soluble hydrate $NH_3.H_2O$ and $2NH_3.H_2O$ are isolated. In these molecules ammonia and water are linked by hydrogen bonds. In aqueous solutions also ammonia may be hydrated in a similar manner and we call the same as ($NH_3.H_2O$)

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

The dielectric constant of ammonia is considerably high to make it a fairly good ionising solvent like water.

$$2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

$$K_{-50^0C} = [NH_4^+][NH_2^-] = 10^{-30}$$

$$2H_2O \rightleftharpoons H_3O^+ + OH^-$$

$$K_{25^0C} = [H_3O^+][OH^-] = 10^{-14}$$

Chemical Properties

Action of heat: Above 500°C ammonia decomposes into its elements. The decomposition may be accelerated by metallic catalysts like Nickel, Iron. Almost complete dissociation occurs on continuous sparking.

$$2NH_3 \xrightarrow{>500^0 C} N_2 + 3H_2$$

Reaction with air/oxygen: Ammonia does not burn in air but burns freely in free oxygen with a yellowish flame to give nitrogen and steam.

$$4NH_3 + 3O_2 \rightleftharpoons 2N_2 + 6H_2O$$

In presence of catalyst like platinum, it burns to produce nitric oxide. This process is used for the manufacture of nitric acid and is known as ostwalds process.

$$4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$$

Reducing property: Ammonia acts as a reducing agent. It reduces the metal oxides to metal when passed over heated metallic oxide.

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